

Blood Supply & Nerve Supply of NOSE

Learning objectives:

At the end of the lecture, the students should be able to describe:

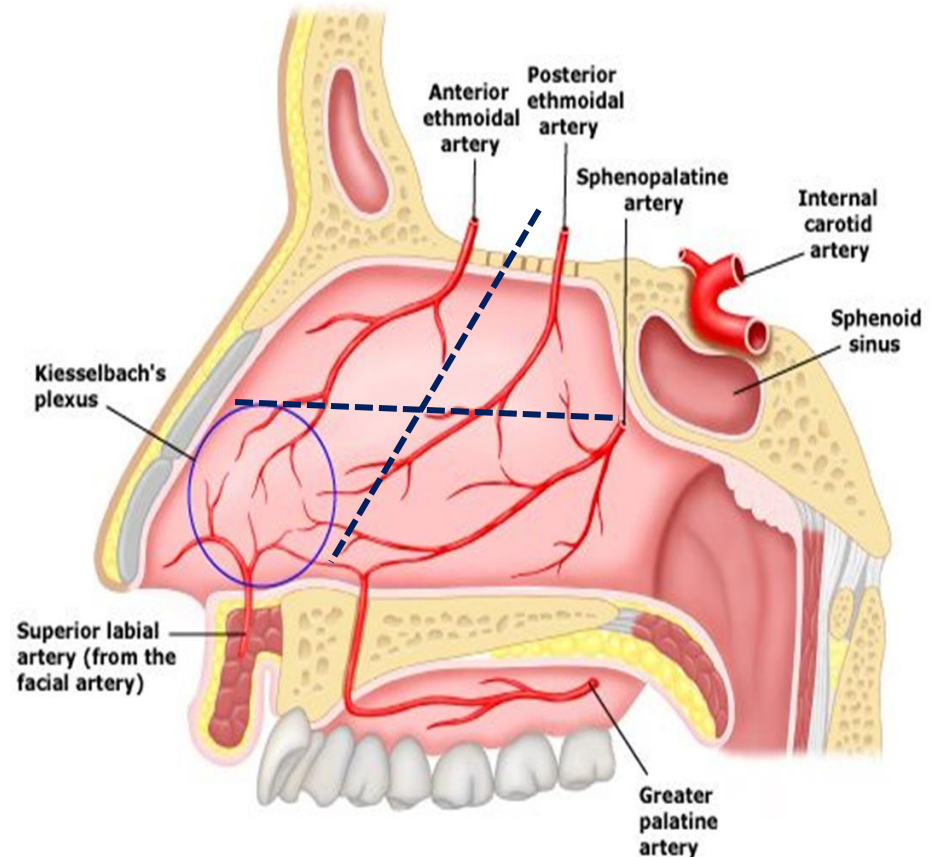
- Arterial supply & venous drainage of nasal septum.
- Little's area, Kiesselbach's plexus & its clinical significance.
- Nerve supply of external nose & nasal septum.
- Arterial supply & venous drainage of lateral wall of nose.
- Nerve supply of lateral wall of nose.

Arterial supply of nasal septum:

- Anterosuperior part: **anterior & posterior ethmoidal arteries**
- Posterosuperior part: **sphenopalatine artery**
- Anteroinferior part: **superior labial branch of facial artery**

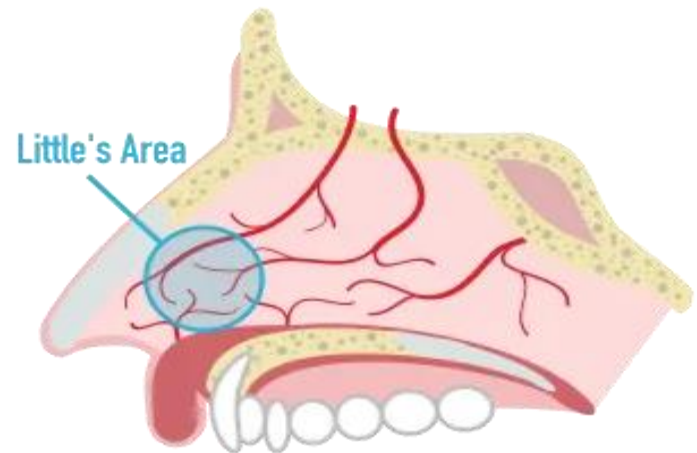
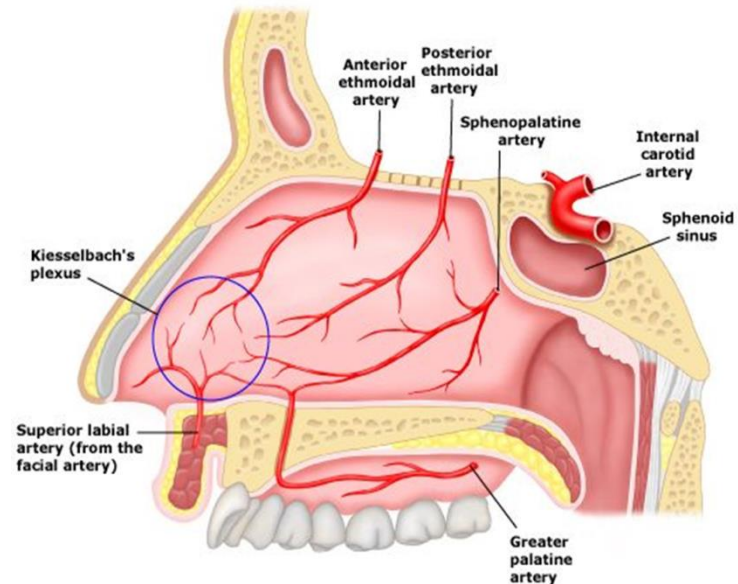
Note:

- Anterior & posterior ethmoidal arteries are the branches of ophthalmic artery.
- Sphenopalatine artery is a branch of maxillary artery.



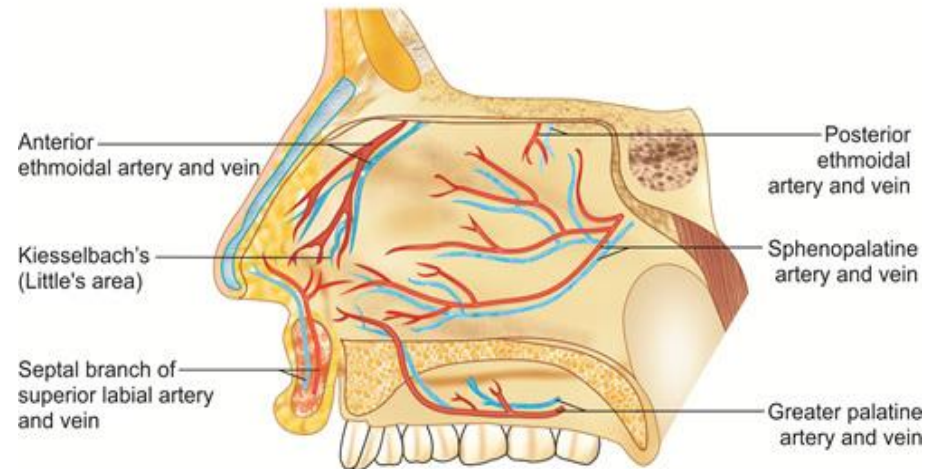
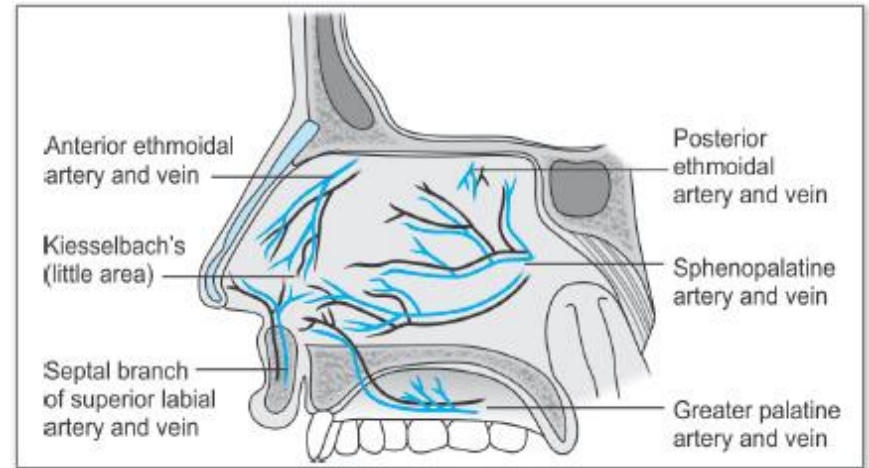
Little's Area & Kiesselbach's plexus:

- The anteroinferior part/vestibule of nasal septum contains anastomoses between superior labial branch of facial artery, branch of sphenopalatine and of anterior ethmoidal artery. This capillary network is called Kiesselbach's plexus.
- **Clinical anatomy:** This is a common site of **epistaxis**/bleeding from nose & is known as **Little's area**, &
- **Sphenopalatine artery** is the main artery of epistaxis.



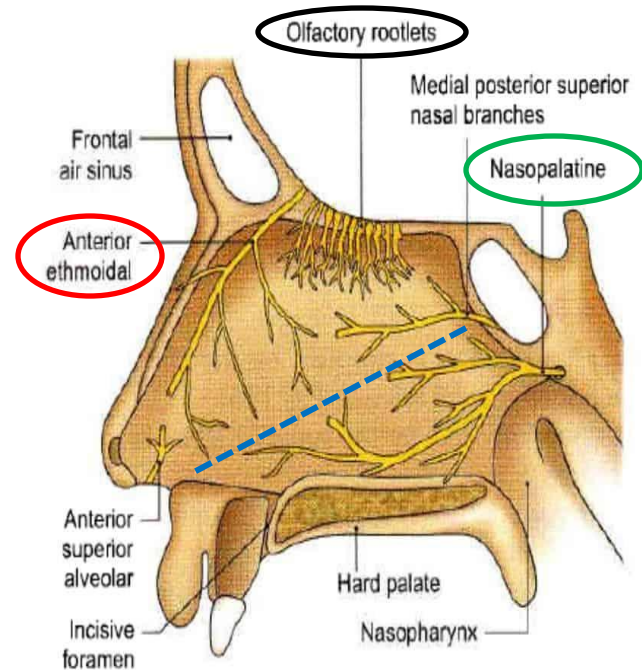
Venous drainage of nasal septum:

- Veins form a plexus which **drains anteriorly** into **facial vein**, &
- Drains posteriorly via sphenopalatine vein into **pterygoid venous plexus**.



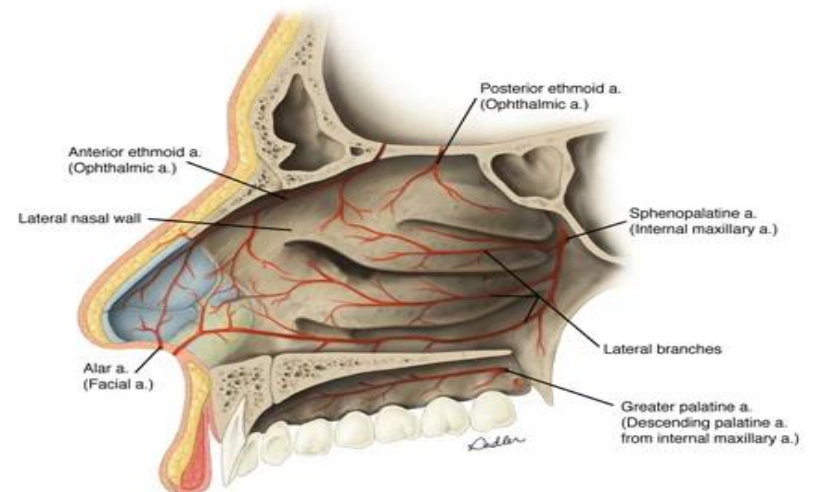
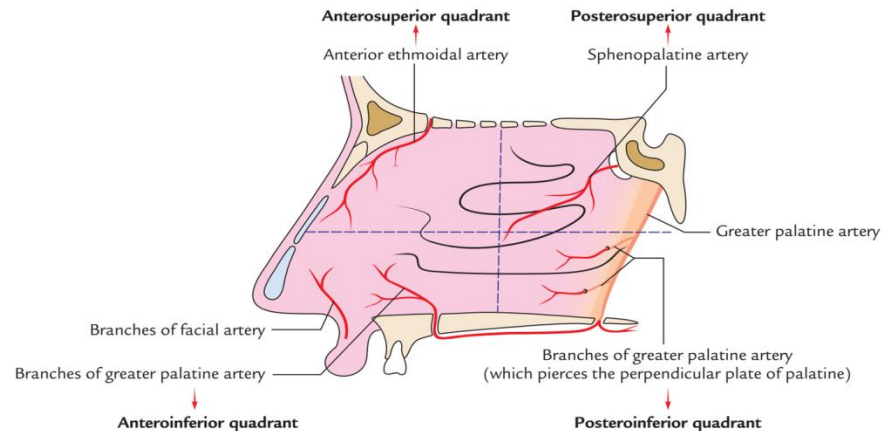
Nerve supply of nasal septum:

- I. **Special sensory:** via **olfactory nerves** confined to the upper part of the nasal cavity.
- II. **General sensory:** via branches of **trigeminal nerve**.
 - Anterosuperior part: **anterior ethmoidal nerve**
 - Posteroinferior part: **nasopalatine branch of pterygopalatine ganglion** (it is the main nerve).

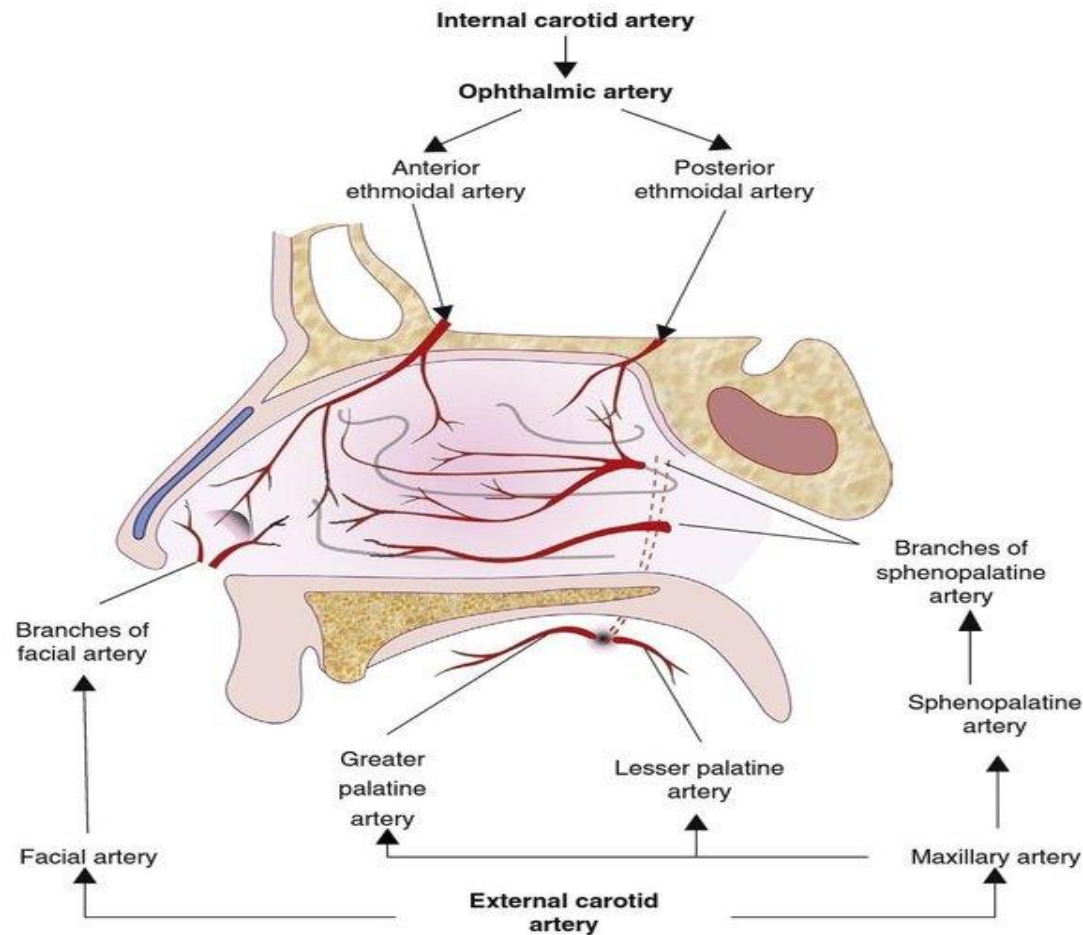


Arterial supply of lateral wall of nose:

- Anterosuperior part: **anterior ethmoidal artery** mainly
- Posterosuperior part: **sphenopalatine artery**
- Anteroinferior part: **facial and greater palatine arteries**
- Posteroinferior part: **greater palatine artery**

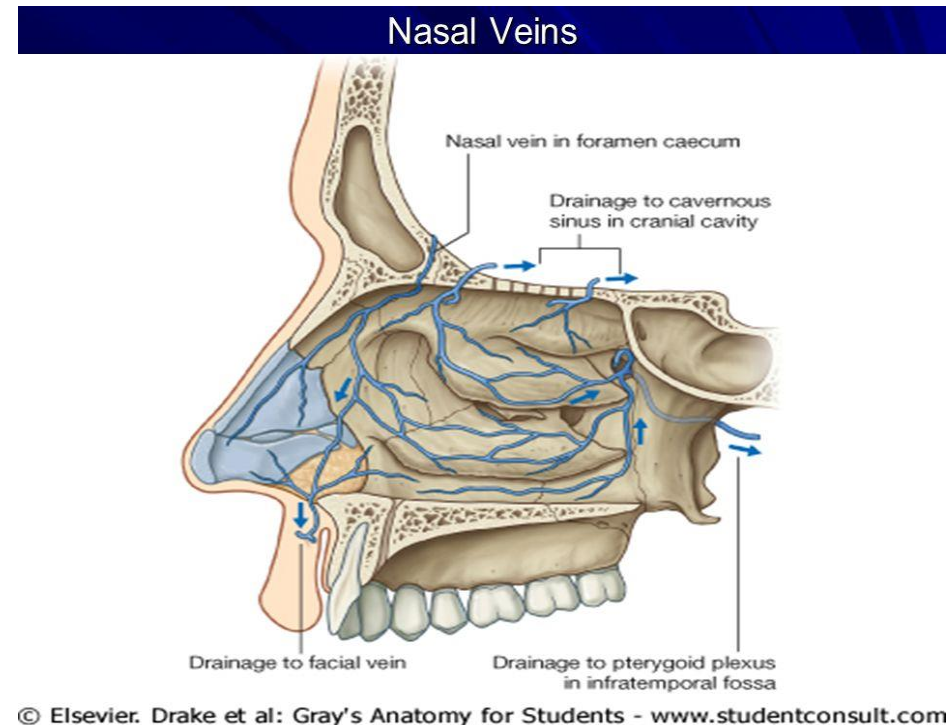


Main sources of arterial supply of nose:



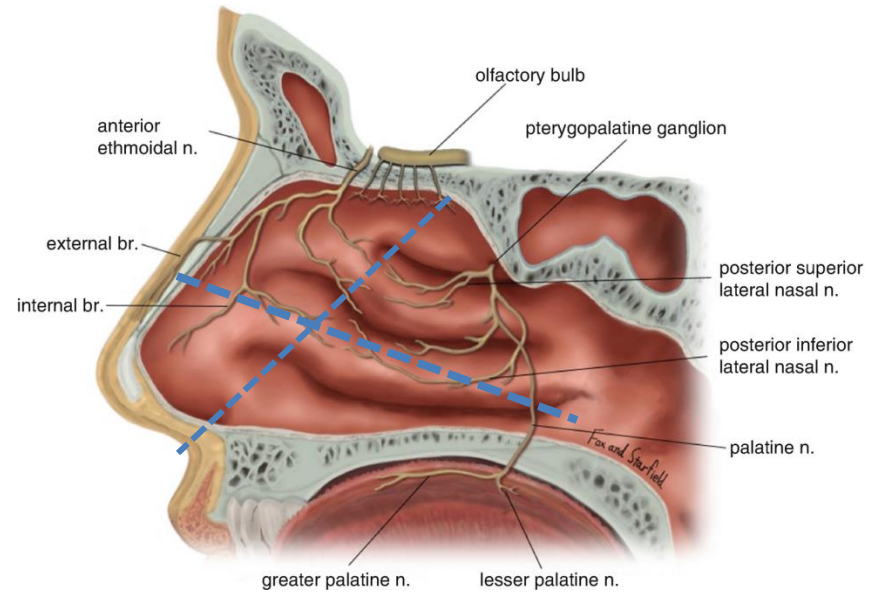
Venous drainage of lateral wall of nose:

- Veins form a plexus which **drains anteriorly into facial vein,**
- Posteriorly into **pharyngeal venous plexus, &**
- Middle part drains into **pterygoid venous plexus**

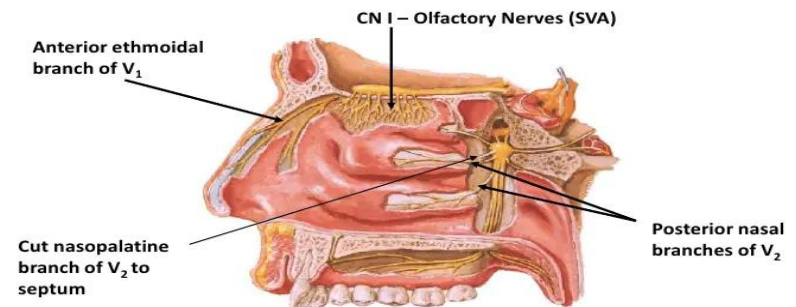


Nerve supply of lateral wall of nose:

- I. **Special sensory:** via **olfactory nerves** are distributed to the upper part, just below cribriform plate (ethmoid) up to superior concha.
- II. **General sensory:** via branches of **trigeminal nerve**:
 - Anterosuperior part: **anterior ethmoidal nerve**
 - Posterosuperior part: **posterior superior lateral nasal branches** of pterygopalatine ganglion (suspended by maxillary nerve)
 - Anteroinferior quadrant: **anterior superior alveolar nerve** (branch of maxillary nerve)
 - Posteroinferior quadrant: **greater palatine branch** of pterygopalatine ganglion (suspended by maxillary nerve)

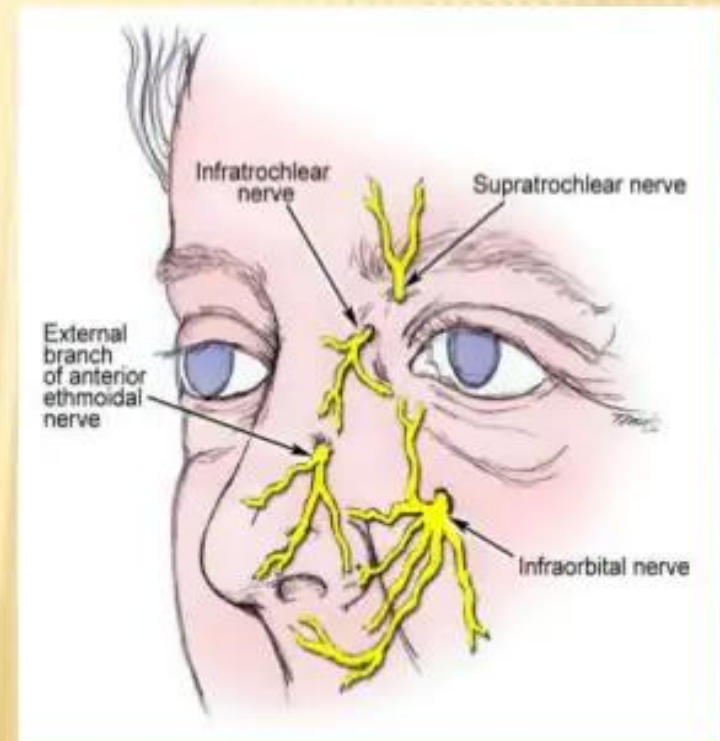


Nerve Supply of the Nasal Cavity



NERVE SUPPLY OF EXTERNAL NOSE

- ✖ Supratrochlear & infratrochlear branches of the ophthalmic nerve supply the skin of root, bridge, upper portion of side of nose
- ✖ Infraorbital branch of the maxillary nerve supply the skin on side of nearly the lower half of nose
- ✖ Ext.nasal branch of Ant. Ethmoidal nerve skin over dorsum to tip



Thank You